Block-Gaussian-Mixture Priors for Hyperspectral Denoising and Inpainting

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***Abstract*— This article proposes a denoiser for hyperspectral (HS) images that consider, not only spatial features, but also spectral features. The method starts by projecting the noisy (observed) HS data onto a lower dimensional subspace and then learns a Gaussian mixture model (GMM) from 3-D patches or blocks extracted from the projected data cube. Afterward, the minimum mean squared error (MMSE) estimates of the blocks are obtained in closed form and returned to their original positions. Experiments show that the proposed algorithm is able to outperform other state-of-the-art methods under Gaussian and Poissonian noise and to reconstruct high-quality images in the presence of stripes.**

***Index Terms*— Expectation–maximization (EM), Gaussian mixture model (GMM), hyperspectral (HS) destriping, image denoising.**

I. Introduction

HYPERSPECTRAL images (HSIs) are very rich sources of information, with a diverse range of applications, such as surveillance, agriculture, and remote sensing [1]. What characterizes HSI is their spectral resolution, that is, instead of having the three standard RGB channels, they have hundreds of bands, each covering a narrow range of the electromagnetic spectrum, giving HSI the potential to identify or discriminate, different land types or materials [2], [3].

In order to achieve a high spectral resolution, the bandwidth of each channel must be small, meaning that few photons reach each sensor and, thus, the observations are severely corrupted by Poisson noise. The presence of noise in HSI thus constitutes a real problem as it may hinder the performance in the applications mentioned earlier. Initial approaches to hyperspectral (HS) denoising handled each band independently, using some state-of-the-art denoiser for grayscale images, such as BM3D [4] or K-singular value decomposition (SVD) [5], but the results were far from what could be obtained because the spectral information was completely ignored (see [6] for a comparison). Subsequent methods attempt to decorrelate the spatial from the spectral information and employ fixed basis transforms, such as the discrete Fourier transform (DFT) or 2-D wavelet transform, and thresholding to recover the clean HS data [7], [8] or applying 3-D filtering strategies [9], [10]. Over the last decade, the paradigm has shifted to spatial and spectral regularization methods, using sparsity-inducing norms or (cubic) total variation [11]–[13], and to methods based on low-rank modeling techniques that leverage the redundancy of the spectral information by exploiting the high correlation between the bands and representing them in a subspace of lower dimension [6], [13]–[16]. For a more comprehensive review, we refer the reader to [17].

In this article, we tackle this important issue limiting the amount of information extracted from HS data, following closely the approach in [6], but replacing the use of a spatial (2-D) denoiser at each band independently, by a 3-D generative model. In particular, our main contribution is a denoising method that learns a Gaussian mixture model (GMM) directly from the subspace representation of the noisy data, using 3-D blocks, instead of the traditional 2-D patches, as in [18]. The rationale is that such an approach has the advantage of capturing simultaneously the interband and intraband correlations, respectively, spectral and spatial features, using the same statistical model. Afterward, we show that the same model can also be successfully applied to handle stripes and deadlines, which are very common in HSIs.

The organization of this article is as follows. Section II formulates the HS denoising/inpainting problems while focusing also on a dimensionality reduction technique used later on. In Section III, we present our main contribution, which is the GMM-based denoiser using 3-D blocks, and in Section IV, we illustrate the proposed method on denoising and inpainting problems. Finally, Section V concludes this article.

II. Problem Formulation

A. HS Denoising

HS denoising can be formulated as any other denoising problem; the difference lies in the data dimensions. Mathematically, the observation model is

$$\mathbf{Y} = \mathbf{Z} + \mathbf{N} \tag{1}$$

where $\mathbf{Y} \in \mathbb{R}^{L \times n}$ denotes the observed (noisy) HSI, with spatial resolution $n = n_1 \times n_2$, and L denotes spectral channels. Similarly, $\mathbf{Z}$ stands for the underlying clean HSI

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we wish to infer, and $\mathbf{N}$ denotes the noise. The latter is typically assumed to be Gaussian independent and identically distributed (i.i.d.), but this is not always the most appropriate approach, particularly when dealing with HSIs. It is often the case that, in an HSI, the noise level varies from channel to channel, and thus, a preprocessing stage may be necessary. In this article, three different noise settings will be considered, following [6].

1) Gaussian i.i.d., where for band j of pixel i, $n_{ij} \sim \mathcal{N}(0, \sigma^2)$, or equivalently for the spectral vector $\mathbf{n}_i \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_L)$.
2) Gaussian non-i.i.d., that is, for pixel i, $\mathbf{n}_i \sim \mathcal{N}(0, \mathbf{D})$, where $\mathbf{D}$ is a diagonal matrix with entries sampled from the uniform distribution $U(0, 1)$ such that each channel has a different noise variance.
3) Poisson noise, such that, rather than (1)

$$\mathbf{Y} \sim \text{Pois}(\alpha \mathbf{Z}) \tag{2}$$

with α such that SNR $= 15$ dB. For Poisson observations of the form (3), the SNR is given by

$$\text{SNR} = 10 \log_{10} \left(\alpha \left(\sum_i Z_i^2 \right) / \sum_i Z_i \right) \tag{3}$$

as shown in the Appendix.

One of the difficulties in the HS denoising problem is its high dimensionality. Exploiting the fact that the bands tend to be highly correlated, this difficulty may be mitigated by assuming that the spectra in $\mathbf{Z}$ live in a low-dimensional subspace, say L_s-dimensional, with $L_s \ll L$. The low dimensionality of the spectra may be seen as resulting from the number of different materials in a scene often being much smaller than the number of bands [2]. This assumption allows shifting the problem from estimating L-dimensional spectra to estimating L_s-dimensional representations of these spectra. Several methods have been proposed to identify a suitable subspace, such as principal component analysis (PCA), SVD, and Hysime [19]. In this article, we adopt Hysime [19], which estimates both the dimension L_s and a basis of the subspace and is completely parameter-free. Letting $\mathbf{E} = [\mathbf{e}_1, \ldots, \mathbf{e}_{L_s}] \in \mathbb{R}^{L \times L_s}$ contain a basis of this subspace, we may write $\mathbf{Z} = \mathbf{E}\mathbf{X}$ (and thus $\mathbf{Y} = \mathbf{E}\mathbf{X} + \mathbf{N}$), where each column of $\mathbf{X} \in \mathbb{R}^{L_s \times n}$ is a representation (vector of coefficients) of the corresponding column of $\mathbf{Z}$. Without loss of generality, we assume that $\mathbf{E}$ is semiunitary such that $\mathbf{E}^T\mathbf{E} = \mathbf{I}_{L_s}$. Then, instead of estimating $\mathbf{Z}$ directly, the problem is reformulated as that of obtaining an estimate $\widehat{\mathbf{X}}$, from which an estimate $\widehat{\mathbf{Z}} = \mathbf{E}\widehat{\mathbf{X}}$ can be obtained.

Algorithm 1 sketches in rough lines the four steps that compose our HS denoising method. In essence, it is similar to the one proposed in [6], but what distinguishes our approach is the type of denoiser that is used. While in standard imaging problems, we estimate the underlying clean image by resorting to priors or regularizers that act on the spatial features, with HSIs, as mentioned earlier, better results can be obtained by leveraging also the rich spectral information [15], [20], [21]. In this article, we show how a GMM-based denoiser can be applied along both spatial and spectral dimensions of the data, thus improving the final restoration results.

Algorithm 1 HS Denoising

Input: Noisy HS data $\mathbf{Y}$;
Output: Clean HS data $\widehat{\mathbf{Z}}$;
1 Estimate/whiten noise $\mathbf{D}$;
2 Learn the subspace $\mathbf{E}$ via HySime [19] or SVD;
3 $\widehat{\mathbf{X}} \leftarrow denoise(\mathbf{Y}, \mathbf{E}, \mathbf{D})$;
4 Reconstruct HS image: $\widehat{\mathbf{Z}} = \mathbf{E}\widehat{\mathbf{X}}$;

Algorithm 2 HS Inpainting

Input: Noisy incomplete data $\mathbf{Y}$;
Output: Clean and complete HS data $\widehat{\mathbf{Z}}$;
1 Determine inpainting masks $\mathbf{M}_i$, $i = 1, \ldots, n$;
2 Learn the subspace $\mathbf{E}'$ from completely observed pixels;
3 Compute the coefficients for incomplete pixels via (4);
4 Obtain complete noisy pixels: $\mathbf{y}_i = \mathbf{E}'\mathbf{x}_i$;
5 Compute $\widehat{\mathbf{Z}}$ via Algorithm 1;

B. HS Inpainting

Besides the presence of noise, HSIs are often affected by stripes or deadlines due to sensor malfunction or miscalibration. These perturbations are characterized by unreliable measurements covering a large extension along the vertical or horizontal axis of specific bands; in HS inpainting, these stripes are treated as missing data. Here, we follow the same approach as in [6], where the inpainting problem is tackled via a suboptimal approach that starts by estimating the subspace from completely observed pixels, imputes the missing observations via least-squares estimation, and lastly obtains the estimate of the clean image with the block-based GMM-denoiser presented in previous section. The algorithm is presented in Algorithm 2; next, we provide a few details that make up a summary of the method proposed in [6].

In the case of HS inpainting, working in a lower dimensional subspace is of particular importance because we may leverage the high correlation that exists between the neighboring bands to estimate the vectors of coefficients, even with missing observations, as

$$\mathbf{x}_i = \left(\mathbf{E}^T \mathbf{M}_i^T \mathbf{M}_i \mathbf{E} \right)^{-1} \mathbf{E}^T \mathbf{M}_i^T \mathbf{y}_{o,i} \tag{4}$$

which is the least squares solution to the simplified inpainting problem given by

$$\mathbf{y}_{o,i} = \mathbf{M}_i \mathbf{z}_i + \mathbf{n}_{o,i} = \mathbf{M}_i \mathbf{E} \mathbf{x}_i + \mathbf{n}_{o,i}. \tag{5}$$

In (5), the problem is formulated for a single-pixel signature, as indicated by the subscript i; $\mathbf{z}_i \in \mathbb{R}^L$ is the complete signature we aim to infer, $\mathbf{y}_{o,i} \in \mathbb{R}^{L_o}$ denotes the observed pixel signature with missing entries, thus $L_o < L$, $\mathbf{n}_{o,i} \in \mathbb{R}^{L_o}$ is Gaussian noise, and $\mathbf{M}_i \in \mathbb{R}^{L_o \times L}$ is a mask acting on the ith pixel. Naturally, $\mathbf{M}_i$ is a subset of the rows of an identity matrix, with a single 1 per row, such that $\mathbf{M}_i \mathbf{M}_i^T = \mathbf{I}_{L_o} \in \mathbb{R}^{L_o \times L_o}$ and $\mathbf{M}_i^T \mathbf{M}_i \in \mathbb{R}^{L \times L}$ is diagonal with ones and zeros at the observed and missing bands, respectively. By considering that the missing pixels have value 0, it is straightforward to identify their locations via a thresholding operation. Another

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issue of practical importance is that we assume that the number of observed samples where the stripes exist is still larger than the size of the subspace, that is, $L_o > L_s$, to ensure the matrix inverse in (4) is always well-defined.

Once every vector of coefficients has been estimated, we may recover the complete noisy observations and subsequently estimate the clean HS data by solving a denoising problem, as formulated in previous section.

III. GMM-Based Denoiser

The GMM denoiser proposed in this section is an extension of the one presented in [18]. The popularity of GMMs is due to their universal approximation property [22], meaning that with enough components and training data, any continuous density can be approximated up to any tolerance level. In addition, the tractability of the model and the ease of training with the expectation–maximization (EM) algorithm allow the model to be learned directly from noisy patches, which is an important feature for the development of models that are targeted specifically to the input image [23], [24]. These important features motivated us, as well as others, to apply GMM to patch modeling, achieving promising results in a variety of inverse imaging problems, not only on natural images [18], [25], [26] but also on HS data [21], [27].

While in [18], the GMM was learned from noisy 2-D patches, extracted from the gray scale input image, here the main features are along the spectral dimension of the input HSI. The simplest case to consider corresponds to the case of 1×1 "patches," which are actually each individual pixel spectra. Using the formulation in (1), the GMM denoiser assumes that each column of $\mathbf{Z}$, denoted by lowercase $\mathbf{z}$, follows a distribution given by:

$$p(\mathbf{z}) = \sum_{m=1}^{K} \alpha_m \mathcal{N}\big(\mathbf{z}; \boldsymbol{\mu}_m, \mathbf{C}_m\big) \tag{6}$$

where $\alpha_1, \ldots, \alpha_K$ are the mixing weights and $\mathcal{N}(\cdot; \boldsymbol{\mu}, \mathbf{C})$ denotes a Gaussian density of mean $\boldsymbol{\mu}$ and covariance matrix $\mathbf{C}$. Such scenario coincides with the standard GMM-based denoiser as presented in [18], which we repeat here for completeness.

Once again, in this article, the noise is considered to be Gaussian (possibly non-i.i.d.) with diagonal covariance matrix $\mathbf{D}$, either known *a priori* or estimated from the observed HSI; the case where the noise is Poissonian can be converted to Gaussian using the Anscombe transform [28]. Because the sum of two Gaussians is a still a Gaussian with covariance equal to the sum of the covariances, the marginal $p(\mathbf{y})$ is

$$p(\mathbf{y}) = \sum_{m=1}^{K} \alpha_m \mathcal{N}\big(\mathbf{y}; \boldsymbol{\mu}_m, \mathbf{C}_m + \mathbf{D}\big). \tag{7}$$

By Bayes' law, $p(\mathbf{z}|\mathbf{y}) = p(\mathbf{y}|\mathbf{z})p(\mathbf{z})/p(\mathbf{y})$, that is

$$p(\mathbf{z}|\mathbf{y}) = \frac{1}{p(\mathbf{y})} \sum_{m=1}^{K} \alpha_m \, p(\mathbf{y}|\mathbf{z}) \mathcal{N}\big(\mathbf{z}; \boldsymbol{\mu}_m, \mathbf{C}_m\big) \tag{8}$$

$$= \frac{1}{p(\mathbf{y})} \sum_{m=1}^{K} \alpha_m \, p_m(\mathbf{y}) \frac{p(\mathbf{y}|\mathbf{z}) \mathcal{N}\big(\mathbf{z}; \boldsymbol{\mu}_m, \mathbf{C}_m\big)}{p_m(\mathbf{y})} \tag{9}$$

where $p_m(\mathbf{y}) = \mathcal{N}(\mathbf{y}; \boldsymbol{\mu}_m, \mathbf{C}_m + \mathbf{D})$. Resorting to the well-known EM algorithm, using the noisy samples, we are able to estimate the parameters of the mixture, namely, the E-step amounts to computing

$$\overline{w}_{ij} = \widehat{\alpha}_j \mathcal{N}\big(\mathbf{y}_i; \widehat{\boldsymbol{\mu}}_j, \widehat{\mathbf{C}}_j + \mathbf{D}\big), \quad w_{ij} = \frac{\overline{w}_{ij}}{\sum_{l=1}^{k} \overline{w}_{il}} \tag{10}$$

where $\widehat{\alpha}_j$, $\widehat{\boldsymbol{\mu}}_j$, and $\widehat{\mathbf{C}}_j$, for $j = 1, \ldots, k$, are the current parameter estimates. In the M-step, the update equations for the α_j and $\boldsymbol{\mu}_j$ parameters are

$$\widehat{\alpha}_j \leftarrow \frac{1}{N} \sum_{i=1}^{N} w_{ij}, \quad \widehat{\boldsymbol{\mu}}_j \leftarrow \frac{1}{N\widehat{\alpha}_j} \sum_{i=1}^{N} w_{ij} \mathbf{y}_i \tag{11}$$

whereas the covariances are updated according to

$$\widehat{\mathbf{C}}_j \leftarrow \mathbf{V}_j \big(\boldsymbol{\Lambda}_j\big)_+ \mathbf{V}_j^T \tag{12}$$

where $\boldsymbol{\Lambda}_j$ is the diagonal matrix with the eigenvalues of the matrix

$$\Sigma_j = \frac{1}{N\widehat{\alpha}_j} \sum_{i=1}^{N} w_{ij} \big(\mathbf{y}_i - \widehat{\boldsymbol{\mu}}_j\big)\big(\mathbf{y}_i - \widehat{\boldsymbol{\mu}}_j\big)^T - \mathbf{D}$$

and $\mathbf{V}_j$ is the matrix with the corresponding eigenvectors. Finally, $(\cdot)_+$ denotes the elementwise positive part operator, that is, $z_+ = \max\{z, 0\}$.

Then, the minimum mean squared error (MMSE) estimator of $\mathbf{z}$ is the posterior expectation, $\hat{\mathbf{z}} = \mathbb{E}[\mathbf{z}|\mathbf{y}] = \int \mathbf{z} p(\mathbf{z}|\mathbf{y}) d\mathbf{z}$, thus

$$\hat{\mathbf{z}} = \frac{1}{p(\mathbf{y})} \sum_{m=1}^{K} \alpha_m \, p_m(\mathbf{y}) \underbrace{\int \mathbf{z} \frac{p(\mathbf{y}|\mathbf{z}) \mathcal{N}\big(\mathbf{z}; \boldsymbol{\mu}_m, \mathbf{C}_m\big)}{p_m(\mathbf{y})} d\mathbf{z}}_{\hat{\mathbf{z}}_m}. \tag{13}$$

Each $\hat{\mathbf{z}}_m$ is the mmse estimate of $\mathbf{z}$ from $\mathbf{y}$, using a single Gaussian prior (the mth component of the GMM). This is well known to be $\hat{\mathbf{z}}_m = (\mathbf{C}_m^{-1} + \mathbf{D}^{-1})^{-1}(\mathbf{D}^{-1}\mathbf{y} + \mathbf{C}_m^{-1}\boldsymbol{\mu}_m)$, finally yielding

$$\hat{\mathbf{z}} = \frac{\sum_{m=1}^{K} \alpha_m \mathcal{N}\big(\mathbf{y}; \boldsymbol{\mu}_m, \mathbf{C}_m + \mathbf{D}\big) \hat{\mathbf{z}}_m}{\sum_{m=1}^{K} \alpha_m \mathcal{N}\big(\mathbf{y}; \boldsymbol{\mu}_m, \mathbf{C}_m + \mathbf{D}\big)} \tag{14}$$

or using compact notation

$$\hat{\mathbf{z}} = \sum_{m=1}^{K} \beta_m(\mathbf{y}) \mathbf{v}_m(\mathbf{y}) \tag{15}$$

where

$$\mathbf{v}_m(\mathbf{y}) = \big(\mathbf{C}_m^{-1} + \mathbf{D}^{-1}\big)^{-1} \big(\mathbf{D}^{-1}\mathbf{y} + \mathbf{C}_m^{-1}\boldsymbol{\mu}_m\big) \tag{16}$$

and

$$\beta_m(\mathbf{y}) = \frac{\alpha_m \mathcal{N}\big(\mathbf{y}; \boldsymbol{\mu}_m, \mathbf{C}_m + \mathbf{D}\big)}{\sum_{m=1}^{K} \alpha_m \mathcal{N}\big(\mathbf{y}; \boldsymbol{\mu}_m, \mathbf{C}_m + \mathbf{D}\big)}. \tag{17}$$

Note that both β_m and $\mathbf{v}_m$ depend on the noisy samples; consequently, although $\mathbf{v}_m(\mathbf{y})$ is a linear function of $\mathbf{y}$, the mmse estimate is a nonlinear function thereof. In the scenario just presented, which considers 1×1 "patches," there is obviously no overlap between samples, and thus, the final clean estimates are obtained via (15).

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Considering a neighborhood around each pixel has been shown to improve denoising performance significantly on grayscale images, by lowering the variance of each sample estimate [29]. For this reason, we propose considering also a neighborhood of spectral signatures, which we refer to as 3-D blocks. Nonetheless, simply considering a neighborhood around each pixel raises some difficulties as the size of the samples increases linearly with the number of pixels of the patch, and the covariance matrices we wish to infer increase with the square of the number of pixels. As an example, if we consider an HSI with n pixels and L channels or bands, the previous scenario with single pixels uses n samples of size L, to infer a total of K scalars corresponding to the mixing weights, K L-dimensional vectors representing the mean of each component, and K $L \times L$ covariance matrices. For HSIs with around 100 bands, this is not an issue but, as the patch size increases, the computational cost becomes too high, in particular for matrix inversions. Considering a small 3×3 patches and 100 bands, the covariance matrices become of size 900×900, which is not only too expensive in terms of computational cost, but the number of unknowns becomes comparable to the number of samples, increasing also the estimation error.

One way to sidestep these issues would be to use only diagonal covariance matrices, instead of free ones; such an approach is not pursued here. Here, we leverage the flexibility and approximation power of GMMs and use them to characterize the distribution of the lower dimensional subspace representation of the HSI, that is, $\mathbf{X}$. Such a simple modification is very powerful as it mitigates significantly the issues mentioned earlier; since typically $L_s < 20$, patches up to 5×5 can now be used. In practice, this modification brings about a few changes in the training stage, using the EM algorithm, and also on the inference stage described earlier.

Consider now the subspace formulation, $\mathbf{Y} = \mathbf{EX} + \mathbf{N}$. Projecting the observed noisy data leads to a denoising problem in the lower dimensional subspace, that is

$$\mathbf{Y} = \mathbf{EX} + \mathbf{N} \Leftrightarrow \left(\mathbf{E}^T\mathbf{E}\right)^{-1}\mathbf{E}^T\mathbf{Y} = \mathbf{X} + \left(\mathbf{E}^T\mathbf{E}\right)^{-1}\mathbf{E}^T\mathbf{N} \quad (18)$$

which, in turn, corresponds to the denoising problem

$$\mathbf{Y}_s = \mathbf{X} + \mathbf{N} \quad (19)$$

where $\mathbf{Y}_s, \mathbf{X}, \mathbf{N} \in \mathbb{R}^{L_s \times n}$, and the noise is now Gaussian with covariance given by $(\mathbf{E}^T\mathbf{E})^{-1}\mathbf{E}^T\mathbf{D}\mathbf{E}(\mathbf{E}^T\mathbf{E})^{-1}$, simplifying to $\mathbf{E}^T\mathbf{D}\mathbf{E}$ by considering the assumption that $\mathbf{E}$ is a semiunitary matrix. The noise covariance matrix is no longer diagonal, except for the case of i.i.d. noise where $\mathbf{E}^T(\sigma^2\mathbf{I})\mathbf{E} = \sigma^2\mathbf{I}$. Moreover, when considering patches around a given pixel, we stack each individual spectral signature and enlarge the denoising problem by a factor of p, which corresponds to the number of pixels in a patch. The final algorithm tackles the denoising problem

$$\mathbf{Y}_p = \mathbf{X}_p + \mathbf{N} \quad (20)$$

where $\mathbf{Y}_p, \mathbf{X}_p, \mathbf{N} \in \mathbb{R}^{(L_s \times p) \times n}$. Note that stacking the spectral signatures also extends the covariance matrices to capture correlations between the neighboring spectra. Furthermore, stacking the spectral signatures also requires that the noise covariance matrix, $\mathbf{E}^T\mathbf{D}\mathbf{E}$, be repeated for each of them. In (7)–eqreftgrs:eq:betam, $\mathbf{D}$ must be replaced by a block diagonal matrix, $\overline{\mathbf{D}}$, computed as

$$\overline{\mathbf{D}} = \mathbf{I}_p \otimes \left(\mathbf{E}^T\mathbf{D}\mathbf{E}\right) \quad (21)$$

where $\otimes$ denotes the Kronecker product.

In case of overlapping blocks, after computing each mmse estimate, they are returned to their location and combined by an averaging procedure. The simplest way to do so is to use a straight average, which corresponds to solving the optimization problem (see [30])

$$\widehat{\mathbf{x}} \in \underset{\mathbf{x} \in \mathbb{R}^n}{\operatorname{Argmin}} \sum_{i=1}^{N} \|\hat{\mathbf{x}}_i - \mathbf{P}_i\mathbf{x}\|_2^2 \quad (22)$$

where $\mathbf{x}$ is a vectorized representation of $\mathbf{X}_p$, and $\mathbf{P}_i \in \{0,1\}^{(L_s \times p) \times n}$ is a binary matrix that extracts the ith block (assumed to cover a neighborhood of p pixels) from the image (thus $\mathbf{P}_i^T$ puts the block back into its place) and N is the number of blocks. The solution to (22) is

$$\hat{\mathbf{x}} = \left(\sum_{i=1}^{N} \mathbf{P}_i^T\mathbf{P}_i\right)^{-1} \sum_{i=1}^{N} \mathbf{P}_i^T\hat{\mathbf{x}}_i. \quad (23)$$

Assuming that the blocks are extracted with unit stride and periodic boundary conditions, every pixel spectra belongs to exactly p blocks and $\sum_{i=1}^{N} \mathbf{P}_i^T\mathbf{P}_i = p\mathbf{I}$, thus its inverse is $p^{-1}\mathbf{I}$ and (23) corresponds to averaging the p estimates of each pixel spectra (one for each of the p blocks that contain it).

IV. Experimental Results

In this section, we report the results of the proposed method on the two different applications. First, we tackle HSI denoising to support our claim that learning a GMM to characterize blocks of pixel spectra provides better results than considering other denoisers applied independently on each subspace band. Afterward, we show that the same model can also accurately recover stripes and deadlines.

To evaluate quantitatively the performance of the methods, we follow the same procedure as in [15] for the Washington DC Mall and Pavia University data sets, that is, very low SNR water absorption bands are removed, and the remaining signatures are projected to a subspace of dimension 8, removing most of the noise, and then back to the original dimension. The resulting HS data are considered the ground-truth image, $\mathbf{Z}$. The Indian Pines data set can be obtained from [31], and we followed the procedure described in [32]. Lastly, before adding synthetic noise to the ground-truth images, each band is normalized to the interval $[0, 1]$. This normalization is useful for two major reasons: 1) with images in the range $[0, 1]$, off-the-shelf denoisers may be used, as is the case and 2) the noise variance across all bands is meaningful, under normalization. Nonetheless, the normalization procedure may have an impact on recovering each pixel spectra due to their large variability across the electromagnetic spectrum. We leave this topic as future work as it requires a careful analysis of how this preprocessing affects the learning of the block-GMM.

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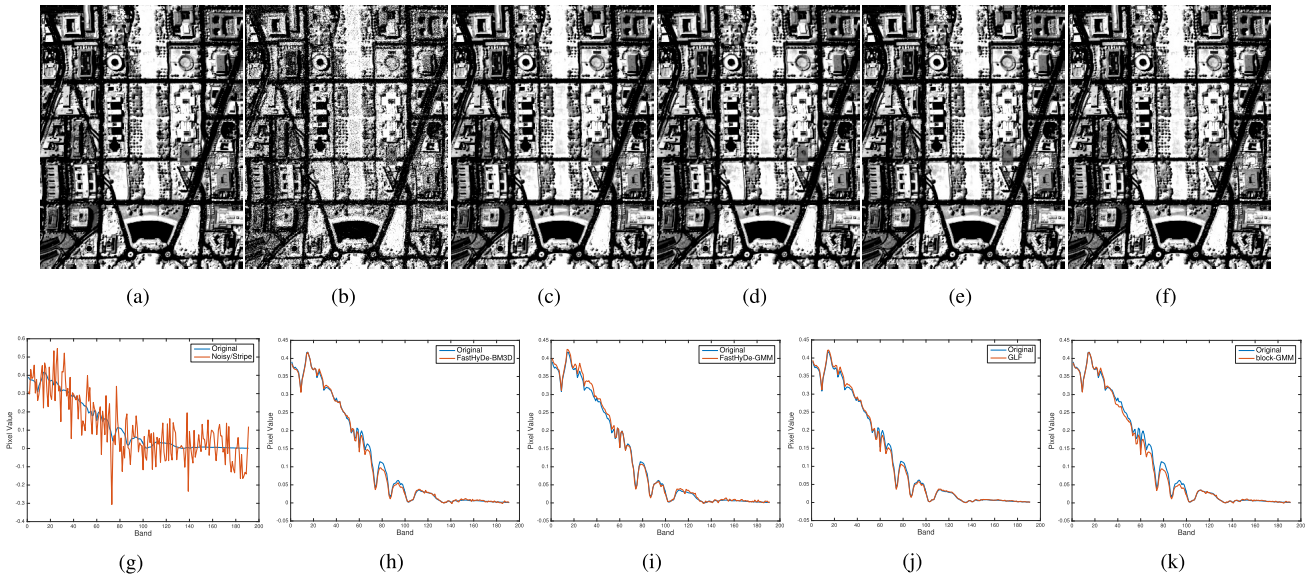


Fig. 1. HS denoising. (a) Original HS band. (b) Noisy HS band (Gaussian i.i.d., with $\sigma = 0.1$). (c) FastHyDe-BM3D. (d) FastHyDe-GMM. (e) GLF. (f) Block-GMM. (g) Random noisy pixel spectra. (h) Restored pixel FastHyDe-BM3D. (i) Restored pixel FastHyDe-GMM. (j) Restored pixel GLF. (k) Restored pixel block-GMM.

TABLE I

RESULTS OBTAINED ON DENOISING UNIVERSITY OF PAVIA DATA SET, FOR DIFFERENT TYPES AND LEVELS OF NOISE. THE BEST RESULT, IN TERMS OF AVERAGE PSNR (TOP), SAM (MIDDLE), AND ERGAS (BOTTOM), IS SHOWN IN BOLD AND THE SECOND BEST IN ITALICS

Noise Type	σ	Input PSNR	FastHyDe BM3D	FastHyDe GMM	GLF	Block-GMM
			47.74	46.59	*48.64*	**49.05**
	0.02	33.98	1.22	1.24	*1.050*	**1.00**
			2.67	2.95	*2.39*	**2.28**
			43.41	43.58	*44.25*	**44.76**
	0.04	27.96	1.84	1.84	*1.60*	**1.51**
			4.37	4.26	*3.92*	**3.69**
			41.04	41.22	*41.76*	**42.34**
Gaussian i.i.d	0.06	24.44	2.29	2.34	*2.03*	**1.89**
			5.72	5.59	*5.19*	**4.84**
			39.40	39.54	*40.03*	**40.54**
	0.08	21.94	2.67	2.76	*2.40*	**2.24**
			6.89	6.77	*6.31*	**5.93**
			38.16	38.26	*38.75*	**39.35**
	0.10	20.00	2.67	2.76	*2.40*	**2.24**
			6.89	6.77	*6.31*	**5.93**
			48.76	45.43	*49.31*	**50.09**
Gaussian non-i.i.d	$U(0,1)$	28.41	1.26	1.49	*1.12*	**1.04**
			2.39	3.33	*2.22*	**2.04**
			43.27	44.10	*45.16*	**45.58**
Poisson	15dB SNR	29.99	1.65	1.63	*1.48*	**1.41**
			3.99	3.88	*3.69*	**3.55**
Avg. Comp. Time	seconds	-	**36.97**	1174.71	*450.03*	508.75

TABLE II

RESULTS OBTAINED ON DENOISING WASHINGTON DC MALL DATA SET, FOR DIFFERENT TYPES AND LEVELS OF NOISE. THE BEST RESULT, IN TERMS OF AVERAGE PSNR (TOP), SAM (MIDDLE), AND ERGAS (BOTTOM), IS SHOWN IN BOLD AND THE SECOND BEST IN ITALICS

Noise Type	σ	Input PSNR	FastHyDe BM3D	FastHyDe GMM	GLF	Block-GMM
			54.63	52.00	*54.95*	**55.40**
	0.02	33.98	0.60	0.70	*0.53*	**0.51**
			4.82	6.99	6.71	*5.34*
			49.98	49.15	*50.27*	**50.75**
	0.04	27.96	1.00	1.01	*0.88*	**0.84**
			7.41	10.05	10.66	*8.28*
			47.19	46.73	*47.35*	**47.89**
Gaussian i.i.d	0.06	24.44	1.32	1.32	*1.16*	**1.10**
			10.66	12.92	14.54	*11.29*
			45.20	44.83	*45.26*	**45.87**
	0.08	21.94	1.60	1.61	*1.42*	**1.34**
			15.40	19.76	20.17	*16.84*
			43.71	43.28	43.52	**44.28**
	0.10	20.00	1.85	1.86	*1.67*	**1.55**
			18.92	24.75	30.84	*21.49*
			56.13	51.37	56.05	**56.36**
Gaussian non-i.i.d	$U(0,1)$	28.62	0.52	0.71	*0.48*	**0.46**
			10.03	18.31	13.78	*13.63*
			44.31	43.65	*46.00*	**46.29**
Poisson	15dB SNR	29.15	1.46	1.44	*1.29*	**1.24**
			3.22	3.49	*3.00*	**2.92**
Avg. Comp. Time	seconds	-	**23.74**	829.79	*293.64*	300.18

In the case of HS images, besides the peak signal-to-noise ratio (PSNR), two other metrics are typically employed to measure the quality of the reconstructed image, which, due to its HS nature, is not meant to be observed by humans. Two examples are the spectral angle mapper (SAM), a metric focused on measuring the quality of the spectral features, and the erreur relative globale adimensionnelle de synthèse (ERGAS), which considers both spatial and spectral features, thus offering a global indication of the quality of the image, for the mathematical details (see [1]).

A. Denoising

Tables I–III show the results on three benchmark data sets, representing urban areas and an agriculture field. For comparison purposes, we use the method proposed in [6], termed FastHyDe, which is a very fast method with close to state-of-the-art results, which tackles each band of the lower dimensional HSI independently of each other, using BM3D [4]. In the spirit of PnP, we also use FastHyDe with a standard GMM denoiser, instead of BM3D, so as to illustrate the difference between modeling the spatial features only or modeling the spatial and spectral features jointly. Finally, we compare the proposed approach with a state-of-the-art method, proposed in [15], which explores both spatial and spectral characteristics of the HSI via low-rank factorization of similar 3-D blocks. Although the selected methods do not cover the wide variety of approaches that have been proposed over the years, these methods achieve results that are at least on the same ballpark as other state-of-the-art methods while sharing most of the building blocks of the proposed approach and having their implementation available, which ensures that the same data are used on every experiment. For

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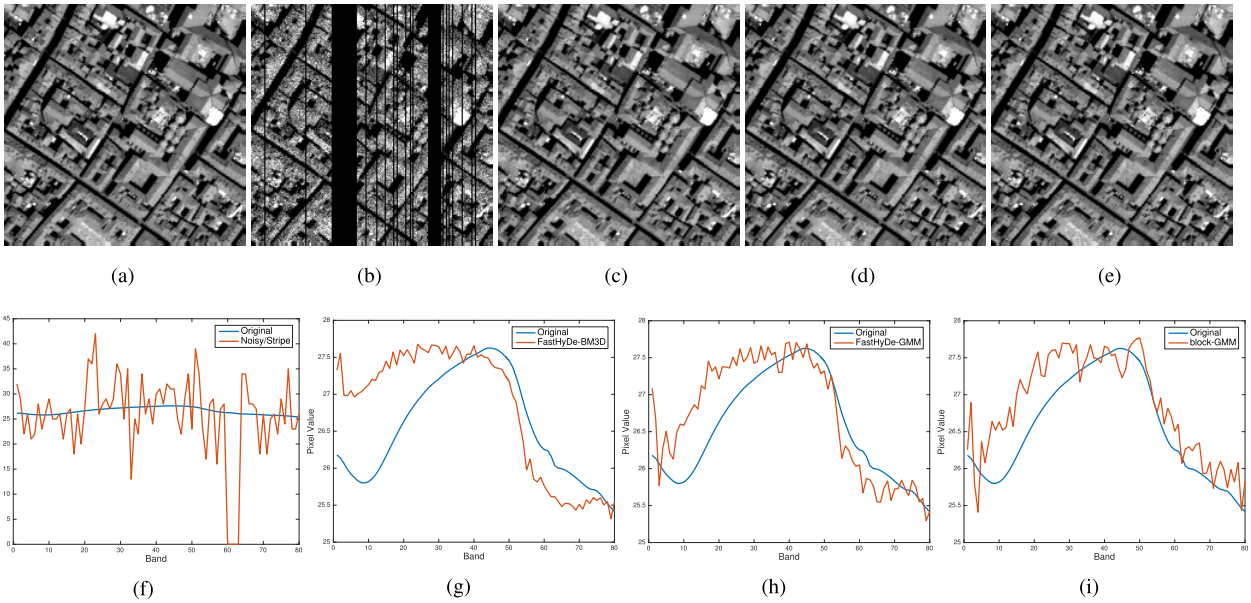


Fig. 2. HS inpainting. (a) Original HS band. (b) HS band with stripes. (c) FastHyDe-BM3D. (d) FastHyDe-GMM. (e) Block-GMM. (f) Random noisy pixel spectra. (g) Restored pixel FastHyDe-BM3D. (h) Restored pixel FastHyDe-GMM. (i) Restored pixel block-GMM.

TABLE III

RESULTS OBTAINED ON DENOISING INDIAN PINES DATA SET, FOR DIFFERENT TYPES AND LEVELS OF NOISE. THE BEST RESULT, IN TERMS OF AVERAGE PSNR (TOP), SAM (MIDDLE), AND ERGAS (BOTTOM), IS SHOWN IN BOLD AND THE SECOND BEST IN ITALICS

Noise Type	σ	Input PSNR	FastHyDe BM3D	FastHyDe GMM	GLF	Block-GMM
Gaussian i.i.d	0.02	33.98	53.28	53.52	*54.85*	**55.86**
			0.56	0.54	*0.50*	**0.47**
			1.36	1.31	*1.12*	**1.01**
	0.04	27.96	49.72	49.62	*50.81*	**51.79**
			0.81	0.82	*0.75*	**0.71**
			2.02	2.05	*1.77*	**1.61**
	0.06	24.44	47.75	47.35	*48.42*	**49.38**
			1.00	1.03	*0.96*	**0.91**
			2.51	2.66	*2.33*	**2.12**
	0.08	21.94	46.38	45.71	*46.86*	**47.51**
			1.17	1.23	*1.15*	**1.09**
			2.93	3.22	*2.78*	**2.61**
	0.10	20.00	45.31	44.48	*45.57*	**46.22**
			1.31	1.41	*1.33*	**1.25**
			3.31	3.71	*3.24*	**3.04**
Gaussian non-i.i.d	$U(0,1)$	28.57	54.32	53.41	54.72	**55.23**
			0.56	0.59	*0.53*	**0.51**
			1.28	1.46	*1.23*	**1.19**
Poisson	15dB SNR	19.41	38.96	38.94	*39.58*	**40.01**
			0.82	0.82	*0.75*	**0.73**
			2.01	1.99	*1.91*	**1.84**
Avg. Comp. Time	seconds	-	**27.73**	558.55	*286.12*	295.39

TABLE IV

RESULTS OBTAINED ON DENOISING WITH STRIPES UNIVERSITY OF PAVIA DATA SET, FOR DIFFERENT TYPES AND LEVELS OF NOISE. THE BEST RESULT, IN TERMS OF AVERAGE PSNR (TOP), SAM (MIDDLE), AND ERGAS (BOTTOM), IS SHOWN IN BOLD AND THE SECOND BEST IN ITALICS

Noise Type	σ	Input PSNR	FastHyIn BM3D	FastHyIn GMM	Block-GMM
Gaussian i.i.d	0.10	19.81	36.96	*37.19*	**37.87**
			4.81	4.92	**4.22**
			7.89	*7.62*	**7.02**
Gaussian non-i.i.d	$U(0,1)$	28.09	*48.50*	45.22	**49.20**
			1.66	1.97	**1.45**
			2.19	2.96	**2.04**
Poisson	15dB SNR	26.51	41.59	*41.81*	**42.78**
			1.63	*1.61*	**1.33**
			3.65	*3.56*	**3.18**
Avg. Comp. Time	seconds	-	**8.90**	133.41	*46.52*

TABLE V

RESULTS OBTAINED ON DENOISING WITH STRIPES WASHINGTON DC MALL DATA SET, FOR DIFFERENT TYPES AND LEVELS OF NOISE. THE BEST RESULT, IN TERMS OF AVERAGE PSNR (TOP), SAM (MIDDLE), AND ERGAS (BOTTOM), IS SHOWN IN BOLD AND THE SECOND BEST IN ITALICS

Noise Type	σ	Input PSNR	FastHyIn BM3D	FastHyIn GMM	Block-GMM
Gaussian i.i.d	0.10	19.90	38.68	*38.83*	**39.40**
			3.27	3.35	**2.95**
			6.95	*6.84*	**6.30**
Gaussian non-i.i.d	$U(0,1)$	28.32	*51.18*	45.96	**51.78**
			1.21	1.36	**1.10**
			1.94	2.83	**1.74**
Poisson	15dB SNR	26.80	*43.24*	42.34	**43.98**
			1.48	*1.46*	**1.30**
			3.23	3.57	**2.97**
Avg. Comp. Time	seconds	-	**20.62**	364.53	*67.97*

HSI denoising, a subspace of dimension 10, that is $L_s = 10$, was used on all the experiments, with basis vectors given by the top ten singular values of the correlation matrix. Moreover, a GMM with 20 components was learned using 3×3 neighborhoods around each pixel, leading to $3\times3\times10$ blocks. The noise variance of each of the bands is estimated using the method from [19].

The results presented in Tables I–III suggest that the proposed method qualitatively outperforms other state-of-the-art methods. Regarding the first two methods, our proposed algorithm was expected to produce higher quality results because, as mentioned earlier, the denoiser works on all the bands jointly instead of dealing with each of them independently. Fig. 1 shows the result on the Washington DC Mall data set with i.i.d. noise, $\sigma = 0.1$, along with the reconstructed spectral signatures of a random missing pixel.

B. Inpainting

As mentioned earlier, besides the noise, it is very common that HS data are corrupted by deadlines or stripes. In this section, we evaluate the performance of the proposed method

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TABLE VI

RESULTS OBTAINED ON DENOISING WITH STRIPES INDIAN PINES DATA SET, FOR DIFFERENT TYPES AND LEVELS OF NOISE. THE BEST RESULT, IN TERMS OF AVERAGE PSNR (TOP), SAM (MIDDLE), AND ERGAS (BOTTOM), IS SHOWN IN BOLD AND THE SECOND BEST IN ITALICS

Noise Type	σ	Input PSNR	FastHyIn BM3D	FastHyIn GMM	Block-GMM
			39.05	*39.17*	**39.46**
Gaussian i.i.d	0.10	19.86	1.46	*1.45*	**1.37**
			4.80	*4.76*	**4.54**
			46.71	45.64	**47.31**
Gaussian non-i.i.d	$U(0,1)$	28.23	0.82	*0.81*	**0.77**
			2.32	2.47	**2.21**
			38.91	*38.94*	**39.93**
Poisson	15dB SNR	19.26	*0.83*	*0.83*	**0.75**
			2.02	*2.00*	**1.86**
Avg. Comp. Time	seconds	-	**42.16**	499.75	*189.44*

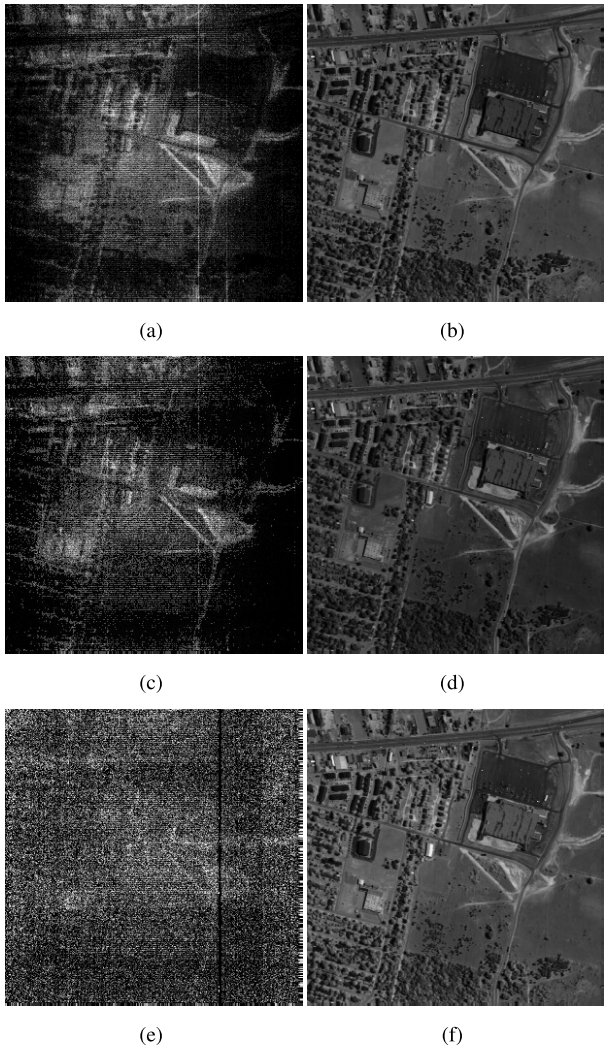


Fig. 3. Urban data set. (a) Original band 139. (b) Restored band 139. (c) Original band 151. (d) Restored band 151. (e) Original band 208. (f) Restored band 208.

(Algorithm 2, with the same parameters as in Section IV-A), against two instances of FastHyIn [6], respectively, with BM3D and the standard GMM-denoiser. Following the same procedure as in [6], we simulate several stripes on bands 60–63. Then, we estimate their locations and feed the information to the algorithm, which reconstructs the missing observations. The results are summarized in Tables IV–VI, for three different types of noise (as before), and they are consistent with the denoising results. In particular, we observe a clear improvement in performance when the reconstruction is done jointly for all images of coefficients than when we treat each image independently. Once again, Fig. 2 shows the result on Pavia University data set with stripes and Poisson noise, along with the reconstructed spectral signatures of a random missing pixel. Yet, since all methods are able to recover the missing observations, visually, it is not obvious which method is best.

C. Real Data Scenario

To illustrate the performance of the proposed method in a real scenario, consider the Urban subscene data set.[1] Some of the bands of this data set are severely corrupted by noise and stripes, as shown in Fig. 3 (left column). Under the assumption of non-i.i.d. noise, estimated with Hysime [19], we run the proposed method with the same parameters as before, that is, a subspace dimension of 10, 20 components in the mixture model, and patches of size 3×3. The results, presented in Fig. 3 (right column), show that the proposed method is able to provide visually pleasing results.

V. CONCLUSION

This article revisited a well-known model for image patches, the GMM, and extended it to handle 3-D patches, called blocks. Such models may be useful for color images, but it is in the HS imaging applications that its improved performance is more pronounced. On the one hand, reducing the HSIs to a lower dimensional space allows the GMM to be trained directly on an accurate representation of the HSI in such a way that all the bands are considered simultaneously, thus exploring interband correlations. On the other hand, by considering a neighborhood of pixels, the learned GMM also accounts for some intraband relations capturing the spatial features. The experiments presented in this article exhibit very promising results, many times surpassing other state-of-the-art methods not only in HSI denoising but also in reconstructing missing information (inpainting).

APPENDIX

For Poissonian observations $\mathbf{Y} \sim \text{Pois}(\alpha \mathbf{Z})$, the SNR is given by (3), as shown next. Note that the signal power is $\alpha^2 \sum_i Z_i^2$ and the noise power is $\mathbb{E}[\sum_i (\alpha Z_i - Y_i)^2]$ (where $\mathbb{E}$ denotes expected value), thus

$$\text{SNR} = 10 \log_{10} \left(\alpha^2 \sum_i Z_i^2 \right) / \mathbb{E} \left[\sum_i (\alpha Z_i - Y_i)^2 \right]. \quad (24)$$

Expanding the denominator, swapping the summation with the expectation and using the fact that $\mathbf{Z}$ is not random

$$\sum_i \mathbb{E}\left[(\alpha Z_i - Y_i)^2\right] = \sum_i \alpha^2 \ Z_i^2 + \mathbb{E}\left[Y_i^2\right] - 2\alpha Z_i \mathbb{E}[Y_i]. \quad (25)$$

[1] Available at: https://rslab.ut.ac.ir/data

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Now, using the fact that $\mathbb{E}[Y_i^2] = \text{var}[Y_i] + \mathbb{E}[Y_i]^2 = \alpha Z_i + \alpha^2\ Z_i^2$ (because $Y_i \sim \text{Pois}(\alpha Z_i)$, we have $\text{var}[Y_i] = \mathbb{E}[Y_i] = \alpha Z_i$)

$$\sum_i \mathbb{E}\left[(\alpha Z_i - Y_i)^2\right] = \sum_i \alpha^2\ Z_i^2 + \alpha Z_i + \alpha^2\ Z_i^2 - 2\alpha^2\ Z_i^2 = \alpha \sum_i Z_i.$$

Plugging this equality into (24) finally leads to (3).

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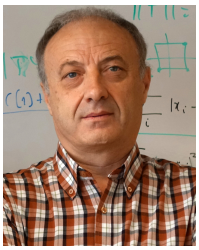


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